

Tyler Kuipers

SOFTWARE ARCHITECT

Edmonton, Canada

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About Me

I've been writing software for about 10 years, with the last three as a backend architect and technical lead. Most of my production work is in Java and JVM (via Gosu, a JVM language used by Guidewire), Python, TypeScript, and Node.js. At Arterys I built the backend services that ran medical AI algorithms on clinical imaging data and returned results to the platform. At AMA I own the API contracts for a platform handling over \$300M in annual transactions, write the RFCs that get the team aligned on hard architectural calls, and set coding standards for a 12-person team. TDD has been part of how I work since 2018: a test framework I designed reduced defects by 30% on subsequent projects, got adopted by 15 developers, and is still running four years later. I treat observability and production reliability as non-negotiable.

Work Experience

Alberta Motor Association

SOFTWARE DEVELOPER IV

Edmonton, Alberta

Jan 2023 - Present

- Architected the infrastructure for Canada's first pay-as-you-go insurance product across multiple AWS accounts. 10 Lambda functions, 99.9% uptime since launch, 2.8M insured trips and 76,637 invoices processed. Won the 2024 Guidewire Innovation Award (1 of 3 worldwide). (Python, TypeScript, AWS Lambda, Terraform)
- Own 4 REST APIs serving 5 internal teams and external partners. Designed the contracts and versioning, approve new endpoint additions, and maintain OpenAPI design standards in Confluence. (Python, TypeScript, REST, OpenAPI)
- Set technical standards for a 12-person team through code reviews, RFC authorship, and PR checklists / API design guides. Authored an RFC moving the team to trunk-based development with feature flags. Worked in cross-functional pods alongside PM, design, and QA. Member of the Architecture Steering Committee, and mentor newer developers through legacy codebases and TDD. (Gosu/Java)
- Built a self-service developer platform solo that took environment provisioning from 2-3 weeks down to 2 hours (60x). Self-service create/destroy, database snapshots, and pull-from-prod, adopted by 15 people across 3 teams and used to spin up 120+ environments. It made production debugging routine: a cancellation bug nobody could reproduce got root-caused in days using snapshotted production data. (Python, Terraform, AWS, Jenkins, Docker)
- Led a 2-3 engineer rewrite of a brittle legacy C# document generation system into containerized ASP.NET Core microservices (.NET 8). 4-5 services on a managed Kubernetes cluster, deployed with Helm. Wired up Datadog metrics for failure rate, throughput, generation time, and storage time. (C#, ASP.NET Core, Kubernetes, Docker, Helm, Datadog)
- Built an event-driven pipeline that ingests an external partner's policy data: S3 ingestion, a Kubernetes CronJob, a translation service, and transactional calls to a composite API, with OAuth 2.0 auth and Datadog monitoring. (Python, Kubernetes, AWS S3, OAuth 2.0)

Arterys Inc. (Acquired by Tempus Labs)

SOFTWARE DEVELOPER

Calgary, Alberta (Remote)

Aug 2021 - Aug 2022

- Built and maintained backend services that ran medical AI algorithms on clinical imaging data and returned results to the platform. (TypeScript, Node.js, AWS)
- Improved AI algorithm processing turnaround time using Redis caching and AWS infrastructure optimizations.
- Monitored production systems via Elasticsearch and Kibana, catching and resolving issues before customer impact.

Alberta Motor Association

SOFTWARE DEVELOPER | SOFTWARE TEAM LEAD

Edmonton, Alberta

Jun 2018 - Feb 2021 | Feb 2021 - Aug 2021

- Designed and built automated test framework that reduced defects by 30% on subsequent projects. Adopted across the insurance engineering team (15 developers / all Guidewire projects) and still in use 4+ years later. (Gosu/Java)
- Led 5-6 person development team on Guidewire insurance platform using TDD methodology. Performed Scrum Master role. Delivered projects on time and budget with no critical defects post-launch. Code reviews and mentoring junior developers.
- Led Git migration for 12-15 developers transitioning from SVN. Created Docker image for local Guidewire development environments. Trained 5-6 developers on TDD methodologies through hosted training sessions.

Pleasant Solutions

SOFTWARE DEVELOPER

Edmonton, Alberta

Jan 2018 - Jun 2018

- Built HMI for a concrete plant control system using TDD methodology. Daily work in C# / .NET, heavily utilizing the MVC design pattern, with REST integration and tests written with NUnit and Moq. (C#, .NET, MVC, REST)

Lockheed Martin Canada

ASSOCIATE SOFTWARE ENGINEER

Halifax, Nova Scotia

Jan 2017 - Jan 2018

- Contributed to Data Fusion module for Combat Management Systems, combining data from multiple underwater and above-water sensor sources. (Java, SQL)
- Designed and implemented scenario recorder system that enabled re-running failed test scenarios at will, improving debugging and testing capabilities.

- Built web app for spatial/temporal data visualization (Edmonton River Atlas). (PHP, JavaScript, PostgreSQL, PostGIS)
- Wrote undergrad thesis on spatial-temporal referencing methods.
- Presented research through poster presentations.

Education

The Open University MSC. COMPUTING (SOFTWARE ENGINEERING) Currently pursuing Master's in Computing at The Open University.	Milton Keynes, England (Remote) May 2024 - Ongoing
The King's University B.S. IN COMPUTER SCIENCE	Edmonton, Alberta Sep 2012 - Dec 2016

Skills

Software Craftsmanship	TDD has been core to how I work since 2018. I designed a test framework that reduced defects by 30% on subsequent projects and got adopted by 15 developers on the insurance engineering team; it's still in use four years later. I care about unit test coverage, continuous integration, and writing code that's easy to change. When a bug gets through, I want to understand why, fix it properly, and make sure it can't happen again.
Backend Services & Ownership	I've spent the last three years as the technical owner of a platform handling over \$300M in annual transactions, building APIs, infrastructure automation, and distributed systems. I own things end to end, from the design doc through the part where they have to stay up. I built a self-service developer platform from scratch that replaced a 2-3 week provisioning process with a 2-hour one, and I architected the backend for a pay-as-you-go product that has maintained 99.9% uptime since launch.
Reliability & Observability	I instrument what I ship with Datadog metrics for failure rate, throughput, and latency, and use them to set clear bars for what healthy looks like. I like chasing hard production failures to root cause: a cancellation bug nobody could reproduce got fixed in days once I could replay real production data against it. To me reliability is part of the feature, not a follow-up.
Technical Leadership & Design Docs	I write the design docs and RFCs that align a team on the hard calls, including an RFC to move us to trunk-based development with feature flags so we could decouple deploys from releases. I set standards through code reviews, PR checklists, and API design guides, sit on AMA's Architecture Steering Committee, and mentor developers through legacy codebases and TDD.

Technologies

Backend Languages	Java / Gosu (JVM), TypeScript, Node.js, Python, C# / .NET (ASP.NET Core, .NET 8), Bash
Architecture & APIs	RESTful APIs, OpenAPI, Microservices, Event-Driven Architecture, API Versioning
Cloud & Infrastructure	AWS (Lambda, EKS, EC2, RDS, S3, SQS, IAM, ECR), Terraform, Kubernetes, Docker, Azure DevOps, Azure Artifacts
Databases	PostgreSQL, SQL, Aurora/RDS, Data Modelling, Redis
AI-Assisted Development	Claude Code, Cursor, GitHub Copilot, Agentic Workflows
Observability	Datadog, Elasticsearch, Kibana, Metrics Design, SLOs, Alerting
CI/CD & Automation	CI/CD, GitHub Actions, Jenkins, GitOps, Feature Flags, Infrastructure as Code
Testing	NUnit, Moq, JUnit, TDD, Test Framework Design